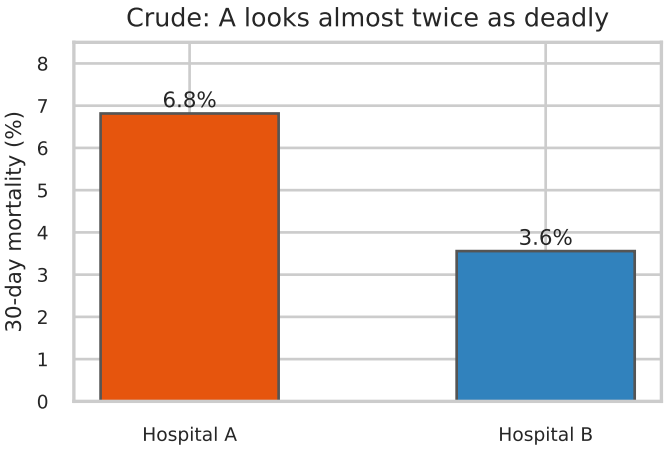
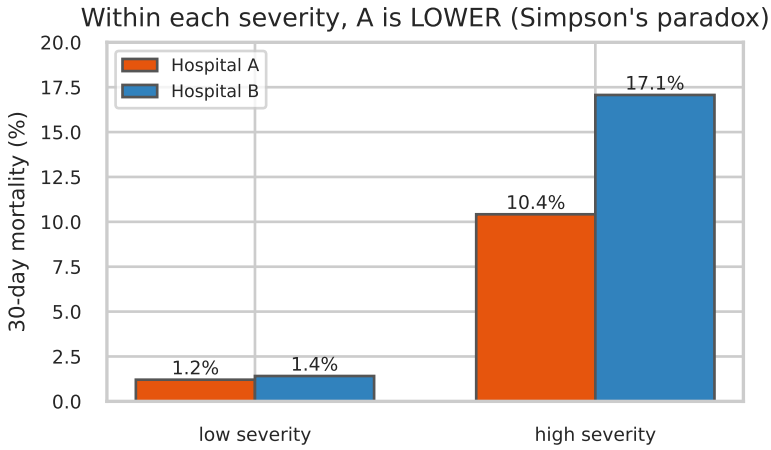
Day 12 Cheat Sheet: Confounding and Risk Adjustment

Comparing hospitals fairly: Simpson's paradox, stratification, standardization, O/E

TERM	WHAT IT MEANS	CODE
Confounder	linked to BOTH the exposure and the outcome	severity -> hospital AND -> death
Crude rate	raw rate, no adjustment	df.groupby("hospital")["died"].mean()
Stratify	compare within levels of the confounder	df.pivot_table(index="severity", ...)
Simpson's paradox	trend in every group reverses when pooled	crude A > B, but A < B in each stratum
Direct standard.	apply stratum rates to ONE standard case mix	(rates * standard_weights).sum()
Mantel-Haenszel	one OR pooled across strata	StratifiedTable(tables).oddsratio_pooled
Breslow-Day	are stratum ORs similar enough to pool?	st.test_equal_odds().pvalue
Adjusted OR	hospital coefficient with confounders in model	smf.logit("died ~ hosp + sev + age")
Expected (E)	sum of case-mix predicted risks	risk_model.predict(df).sum()
O/E ratio	observed / expected; < 1 is better than expected	observed / expected
Risk-adj. rate	O/E times the overall rate	oe * df.died.mean()
Mediator	ON the causal path: don't adjust it away	e.g. a complication after surgery





THIS DATA (simulated, seed 12): 6,000 surgical admissions, two hospitals
Crude 30-day mortality: A 6.8%, B 3.6% | high-severity share: A 61%, B 14%
Within strata: low 1.2% vs 1.4%, high 10.4% vs 17.1% (A lower in both)
Standardized to the combined mix: A 4.6%, B 7.2%
Odds ratio A vs B: crude 1.98, Mantel-Haenszel 0.62 (95% CI 0.47-0.81), logistic 0.60
O/E: A 0.89, B 1.29

HABITS THAT KEEP YOU HONEST
- Before ranking any providers, ask: do they treat the same kind of patients?
- Check the confounder both ways: is it tied to the group AND to the outcome?
- Show crude AND adjusted numbers side by side, and name what you adjusted for
- Adjust for things that exist BEFORE the exposure (severity, age), not after it
- Adjustment only fixes confounders you measured; say so in the limits

```
strat = df.pivot_table(index="severity", columns="hospital", values="died")
std = df["severity"].value_counts(normalize=True); adj = strat.mul(std, axis=0).sum()
df["exp"] = smf.logit("died ~ C(severity) + age", df).fit(disp=0).predict(df)
oe = df.groupby("hospital")["died"].sum() / df.groupby("hospital")["exp"].sum()
```

Infographic: Which Hospital Is Safer?

Simulated: 6,000 surgical admissions, 30-day mortality, two hospitals

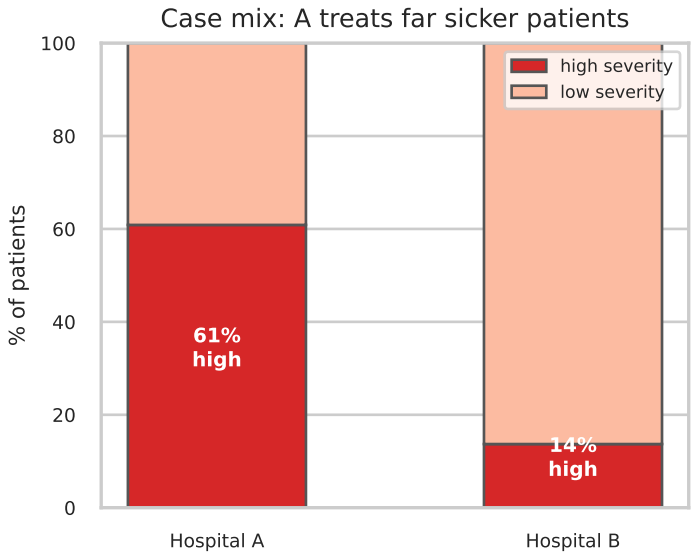
Crude	Mix	Stratify	Standardize	Model	Report
raw rate per group	who does each treat?	compare like with like	one standard case mix	adjusted OR, O/E ratio	crude AND adjusted

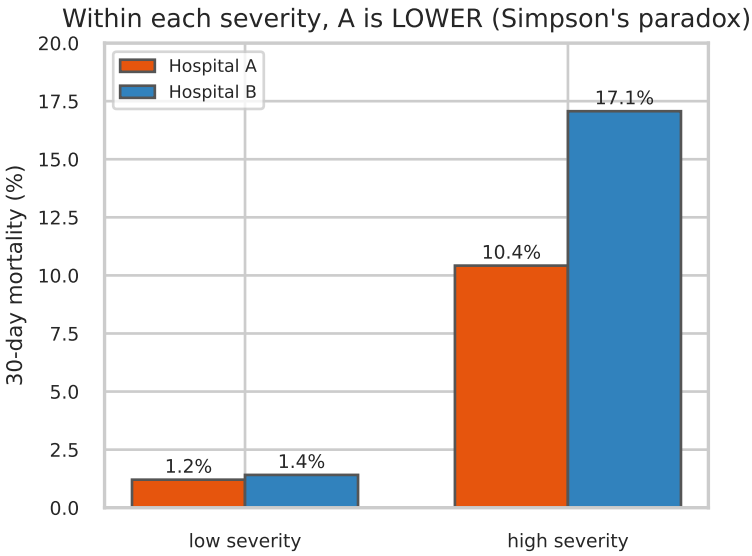
CRUDE
A 6.8% vs B 3.6%
A looks worse

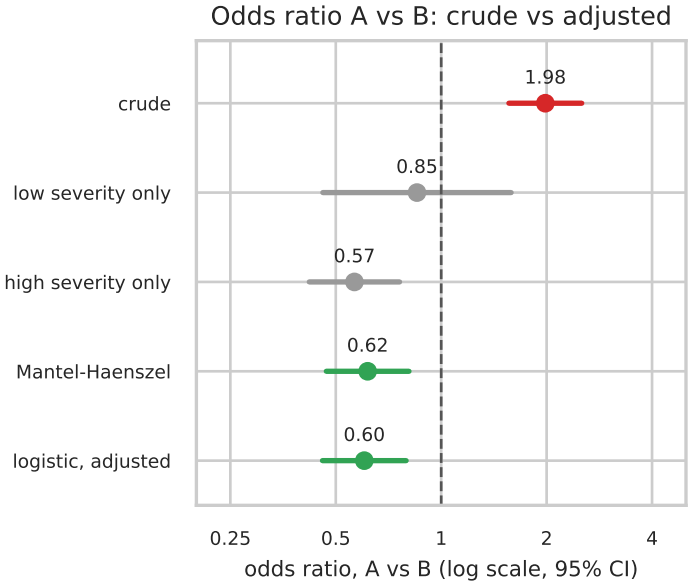
HIGH SEVERITY
A 61% vs B 14%
of patients

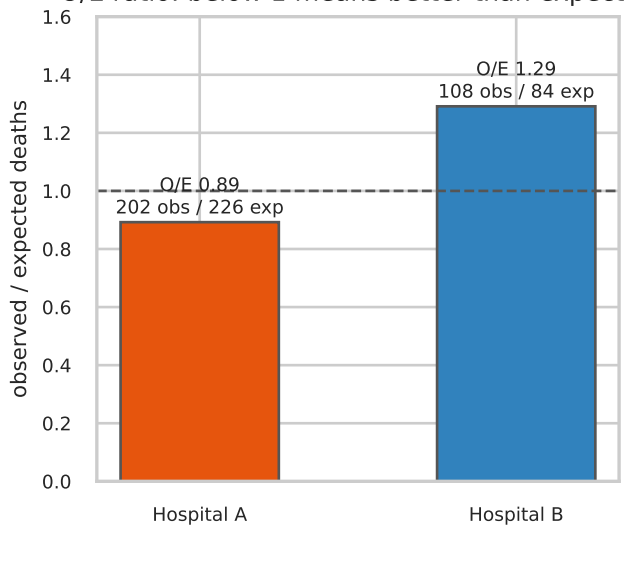
ADJUSTED OR
0.62
A is safer

O/E
A 0.89 vs B 1.29
1.00 = as expected









Read it like a quality director: the referral center gets the sickest patients, so its crude death rate is higher.
Compared like with like, it does better in both severity groups.
Ranking hospitals on crude rates punishes the ones that take hard cases. Simulated data.

Hands-On Worksheet: ds-genai-90-days / day12

60-90 minutes | notebook: day12/confounding_risk_adjustment.ipynb

[ ] mkdir day12, conda activate ds90, open confounding_risk_adjustment.ipynb

[ ] Simulate surgical_outcomes.csv with seed 12 (snippet 1)

[ ] Crude mortality by hospital: write down which one looks worse

[ ] Crosstab hospital x severity: how different is the case mix?

[ ] Stratify: mortality by hospital within each severity (snippet 2)

[ ] Confirm severity is tied to BOTH hospital and death

[ ] Direct standardization to the combined case mix (snippet 3)

[ ] Crude OR vs Mantel-Haenszel OR; check Breslow-Day p

[ ] Logistic regression: crude vs adjusted OR for hospital (snippet 4)

[ ] Case-mix model without hospital; O/E ratio per hospital (snippet 5)

[ ] Plot crude vs adjusted side by side (one chart for a director)

[ ] Findings cell: 5 bullets + limits (unmeasured confounders)

[ ] git add day12 && git commit && git push (snippet 6)

[ ] Stretch: Snowflake SQL O/E by hospital (snippet 7)

```
import pandas as pd, statsmodels.formula.api as smf
df = pd.read_csv("surgical_outcomes.csv")
print(df.groupby("hospital")["died"].mean())             # crude
print(df.pivot_table(index="severity", columns="hospital", values="died"))
m = smf.logit("died ~ hosp_a + C(severity) + age", df).fit(disp=0)
print(np.exp(m.params["hosp_a"]))                         # adjusted OR
```

Crude A: _____ Crude B: _____ High-severity share A: _____ B: _____ Std. A: _____ Std. B: _____

Crude OR: _____ MH OR: _____ Adjusted OR: _____ O/E A: _____ O/E B: _____

Day 12 Self-Check Quiz

1. What two links make severity a confounder here?

_____

2. Hospital A is lower in BOTH strata but higher overall. How is that possible?

_____

3. What does direct standardization hold fixed, and why does that help?

_____

4. Crude OR is about 2.0, Mantel-Haenszel OR about 0.6. Which do you report?

_____

5. What does an O/E ratio of 1.29 mean for Hospital B?

_____

6. Why leave hospital OUT of the expected-risk model?

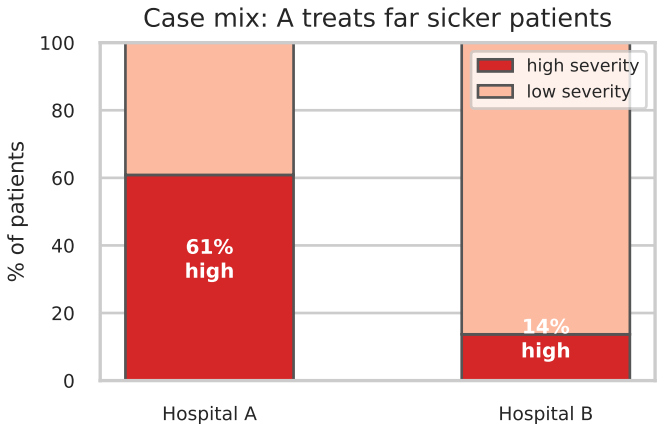
_____

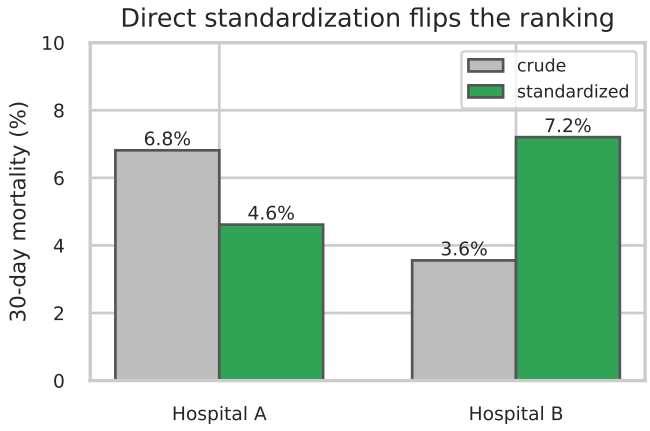
7. Should you adjust for a surgical complication that happened after admission?

_____

8. Adjusted results favor A. What can adjustment still not fix?

_____





Answers:
1 it's tied to the hospital (A 61% high severity vs B 14%) AND to death (high 11.7% vs low 1.3%)
2 the crude rate is a weighted average of stratum rates, and most of A's patients sit in the deadly high stratum (Simpson's paradox)
3 it holds the case mix fixed: both hospitals get the same standard mix, so only stratum rates differ (A 4.6% vs B 7.2%)
4 the adjusted one (0.62, CI 0.47-0.81), shown next to the crude one with a note on what was adjusted for
5 B had 108 deaths where its case mix predicted about 84: roughly 29% more deaths than expected
6 expected means "what an average hospital would see with these patients"; putting hospital in would bake each hospital's own performance into E
7 no: it's on the causal path (a mediator). Adjusting for it hides part of the very quality difference you want to measure
8 confounders you didn't measure (frailty, social factors), differences in how severity is coded, chance; adjusted is better, not proof